

* (From pg 77 of your notes)

$$n t \frac{d\sigma(\theta, \phi)}{d\Omega} d\Omega = \left(\begin{array}{l} \text{Fraction of particles scattered} \\ \text{into a solid angle } d\Omega \end{array} \right)$$

→ At first this may not appear to be what we want as we have no information about n & t however note we want a **ratio**:

$$\frac{\text{ratio of particles scattered forward}}{\text{ratio of particles scattered backwards}}$$

So the n & t will cancel in the ratio.

• Lets look at the above equation again:

$$\frac{dN_{\text{for}}}{N_{\text{tot}}} = n t \frac{d\sigma(\theta, \phi)^f}{d\Omega} d\Omega^f \quad (\text{Forward})$$

I've made the text "Fraction of particles scattered into a solid angle $d\Omega$ "

mathematically explicit, defining new terms

N_{for} : Number of particles scattered forward

N_{tot} : Total number of particles scattered

• You may wonder why the equation has dN_{for} and not just N_{for} : well, recall that the equation is true only for the solid angle $d\Omega$. Thus, we only get dN_{for} for each $d\Omega$.

• I'm doing this for N_{for} but the same applies to

N_{back} :

$$\frac{dN_{\text{back}}}{N_{\text{tot}}} = n t \frac{d\sigma(\theta, \phi)^b}{d\Omega} d\Omega^b$$

Note: I've written $\frac{d\sigma^b}{d\Omega}$ and $\frac{d\sigma^f}{d\Omega}$ to distinguish the two. You'll see why below

To get the total particles in each direction we must integrate

$$\int \frac{dN_{\text{for}}}{N_{\text{tot}}} = n_t \int \frac{d\sigma^f}{d\Omega} d\Omega^f$$

where

$$N_{\text{for}} = \int dN_{\text{for}} \quad \left(N_{\text{back}} = \int dN_{\text{back}} \right)$$

Thus,

$$\frac{N_{\text{for}}}{N_{\text{tot}}} = n_t \int \frac{d\sigma^f}{d\Omega} d\Omega^f$$

or

$$\begin{aligned} N_{\text{for}} &= N_{\text{tot}} n_t \int \frac{d\sigma^f}{d\Omega} d\Omega^f \\ N_{\text{back}} &= N_{\text{tot}} n_t \int \frac{d\sigma^b}{d\Omega} d\Omega^b \end{aligned}$$

* Finally, we want: $\frac{N_{\text{back}}}{N_{\text{for}}}$

(Doesn't matter if we find $\frac{N_b}{N_t}$ or $\frac{N_t}{N_b}$, either) is fine.

Using $\frac{d\sigma}{d\Omega} = \frac{\alpha^2 \hbar^2 c^2}{4 E_{\text{cm}}^2} [1 + \cos^2(\theta) + C_{\text{weak}} \cos(\theta)]$

$$\frac{N_{\text{back}}}{N_{\text{for}}} = \frac{N_{\text{tot}} n_t \int \frac{\alpha^2 \hbar^2 c^2}{4 E_{\text{cm}}^2} (1 + \cos^2(\theta) + C_{\text{weak}} \cos(\theta)) d\Omega^b}{N_{\text{tot}} n_t \int \frac{\alpha^2 \hbar^2 c^2}{4 E_{\text{cm}}^2} (1 + \cos^2(\theta) + C_{\text{weak}} \cos(\theta)) d\Omega^f}$$

→ Everything in green cancels out as we said above!

$$\frac{N_{\text{back}}}{N_{\text{for}}} = \frac{\int (1 + \cos^2(\theta) + C_{\text{mean}} \cos(\theta)) d\Omega^b}{\int (1 + \cos^2(\theta) + C_{\text{mean}} \cos(\theta)) d\Omega^f}$$

* In spherical coordinates (in steradians) [Pg 78]

$$d\Omega = \sin(\theta) d\theta d\phi$$

(see image 6.4 pg 77)

$$\begin{array}{l|l} \phi \in [0, 2\pi] & \int d\Omega = \int_0^{2\pi} \int_0^\theta \sin(\theta) d\theta d\phi \\ \theta_{\text{for}} \in [0, \pi/2] & = \left(\int_0^{2\pi} d\phi \right) \left(\int_0^\theta \sin(\theta) d\theta \right) \\ \theta_{\text{back}} \in [\pi/2, \pi] & \end{array}$$

$$\bullet N_{\text{back}} = \left(\int_0^{2\pi} d\phi \right) \int_{\pi/2}^\pi (1 + \cos^2(\theta) + C_{\text{mean}} \cos(\theta)) \sin(\theta) d\theta$$

$$N_{\text{for}} = \left(\int_0^{2\pi} d\phi \right) \int_0^{\pi/2} (1 + \cos^2(\theta) + C_{\text{mean}} \cos(\theta)) \sin(\theta) d\theta$$

Note that since there is no ϕ dependence the integral can be "factored" out. Thus, it will cancel using $\sin(\theta) d\theta = -d\cos(\theta)$.
(note the negatives will cancel as well)

$$\frac{N_{\text{back}}}{N_{\text{for}}} = \frac{\int_{\pi/2}^\pi (1 + \cos^2(\theta) + C_{\text{mean}} \cos(\theta)) d\cos(\theta)}{\int_0^{\pi/2} (1 + \cos^2(\theta) + C_{\text{mean}} \cos(\theta)) d\cos(\theta)}$$

... Second typo ...

Typo: $\frac{\pi}{2}$ NOT π

Thus, all you have to do are each of the two integrals. Pretty straight forward but if you have any problems just let me know!

[Note you'll be left with some numbers and a fraction containing only C_{mean} just as the

Question asks.

* Important: So, I wrote $\frac{d\sigma^b}{d\Omega} d\Omega^b$ and $\frac{d\sigma^f}{d\Omega} d\Omega^f$
the reason for this is, as you saw, each have different values for θ . Now, naively one might take the ratio before we integrate and so some students will do this:

$$\frac{dN_b}{dN_f} = \frac{\frac{d\sigma}{d\Omega} d\Omega}{\frac{d\sigma}{d\Omega} d\Omega}$$

and "cancel" this term which is wrong.

* Hopefully this makes sense to you and if not just let me know!

Happy Holidays!

